Gastric Mucosa — Regional Structure

LG 2
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GIT BLOCK

BEFORE YOU READ — 3 BLIND ID CHALLENGES

1. A field shows shallow gastric pits and long, straight glands packed with pink (acidophilic) cells. Which region, and which cell type dominates?
2. A field shows very deep, long pits and short, coiled glands with almost no pink cells. Which region is this?
3. Name the one cell in the gastric gland responsible for both HCl and intrinsic factor — and what happens if it's lost?

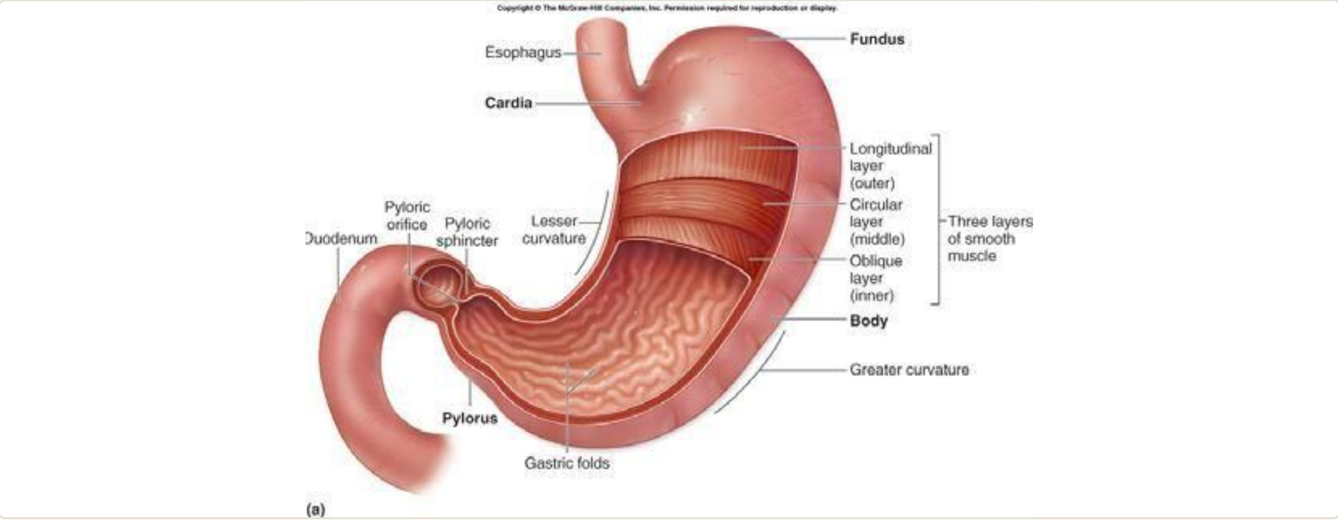
1 · The Four Regions — one-line ID rules

Cardia: a narrow zone at the esophageal opening — shallow pits and short, mostly-mucous glands of roughly equal length.

Fundus & body (histologically identical): the largest zone — shallow pits, long straight glands packed with parietal and chief cells. If you see abundant pink (acidophilic) parietal cells, you're in fundus/body.

Pylorus: distal 4–5cm — deep, long pits with short, coiled glands, mucus-dominant, scattered G cells (gastrin) and D cells (somatostatin).

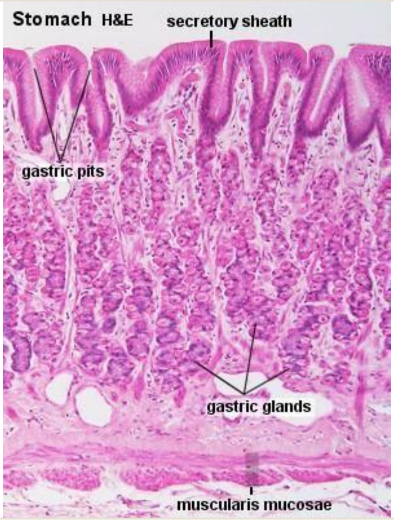
OVERVIEW



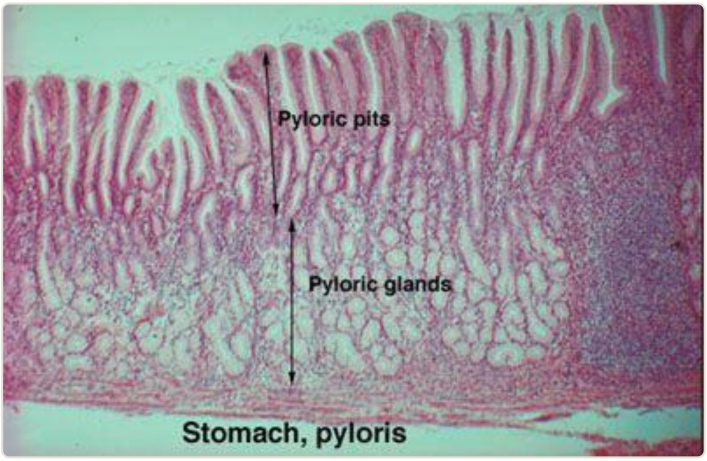
Stomach regions (cardia, fundus, body, pylorus) and the three muscularis externa layers (outer longitudinal, middle circular, inner oblique — thickened at the pylorus to form the pyloric sphincter).

DISCRIMINATOR — THE ONE FINDING THAT SEPARATES EACH REGION

REGION	PIT : GLAND RATIO	DOMINANT CELLS
Cardia	Shallow pits, short glands (roughly equal length)	Mostly mucous cells, few parietal cells
Fundus / body	Shallow pits, **long** glands	Parietal + chief cells abundant (this is the acid/enzyme-producing zone)
Pylorus	**Long/deep** pits, short coiled glands	Mucus-dominant; chief cells scarce; G cells and D cells present



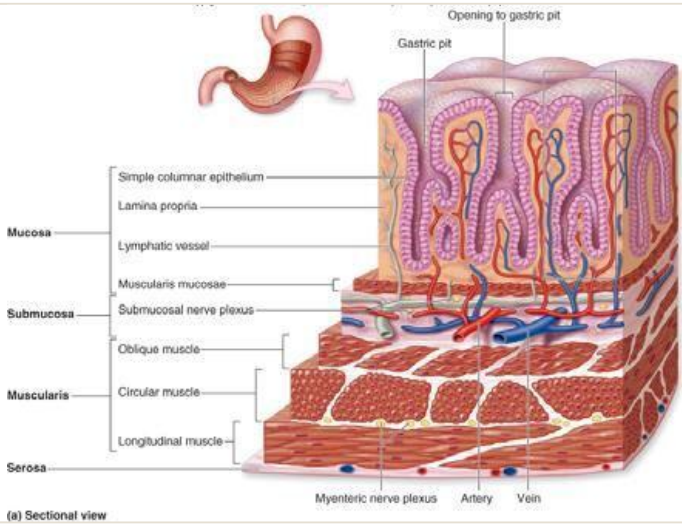
Fundus/body region, low power — shallow gastric pits, long gastric (fundic) glands, muscularis mucosae below.



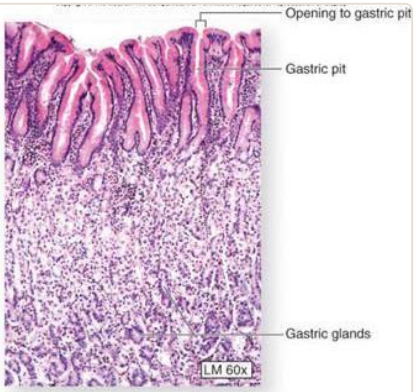
Pyloric region, low power — long pyloric pits, short pyloric glands. Compare the pit:gland proportion against the fundic image.

2 · Mucosal Layers (all regions share this)

LAYER	CONTAINS
a. Epithelium	Simple columnar, invaginates into lamina propria forming **gastric pits**, which empty the cardiac/gastric/pyloric glands. All surface cells secrete mucus + tight junctions = the acid barrier.
b. Lamina propria	Loose CT with smooth muscle, lymphoid cells, glands
c. Muscularis mucosae	Smooth muscle producing local mucosal movement



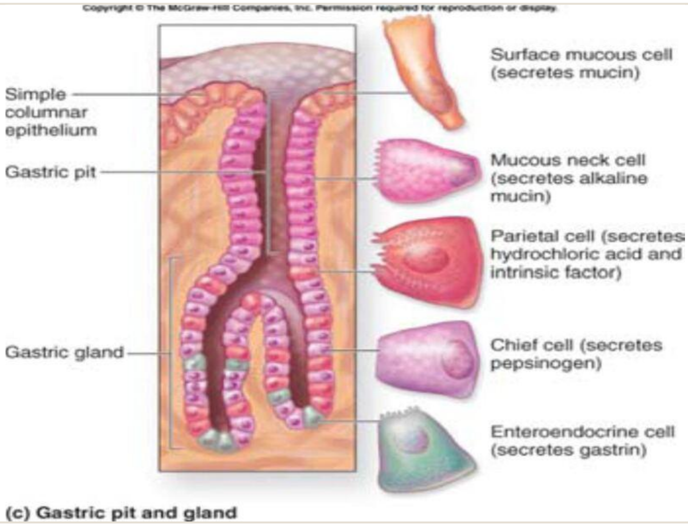
General stomach wall — surface columnar epithelium, gastric pit, lamina propria, muscularis mucosae, submucosa, three-layer muscularis, serosa.



Gastric pit opening into the underlying gastric glands — low power.

3 · The Five Gastric Gland Cell Types

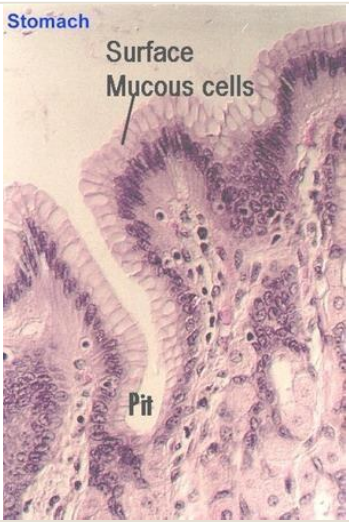
SCAN PATH — FROM ISTHMUS TO BASE



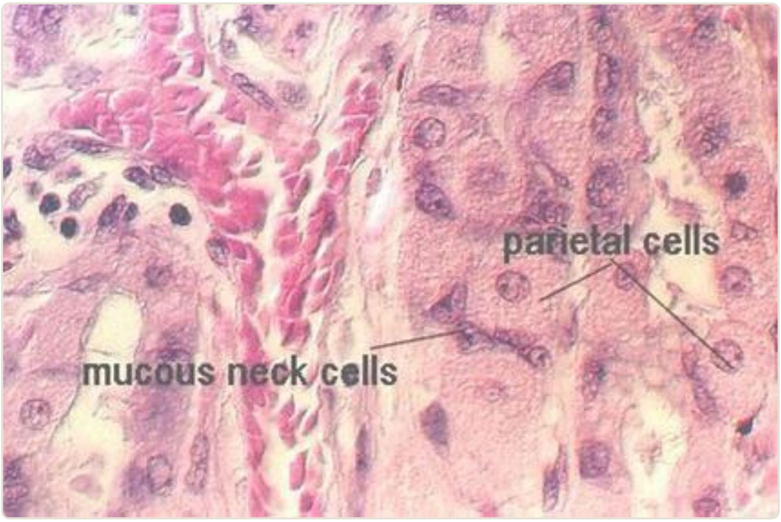
Scan the gland top to bottom: ① isthmus (surface lining + regenerative/stem cells) → ② neck (mucous neck cells, some parietal cells) → ③ base (chief cells predominate, few parietal, scattered enteroendocrine).

STRUCTURE → FUNCTION

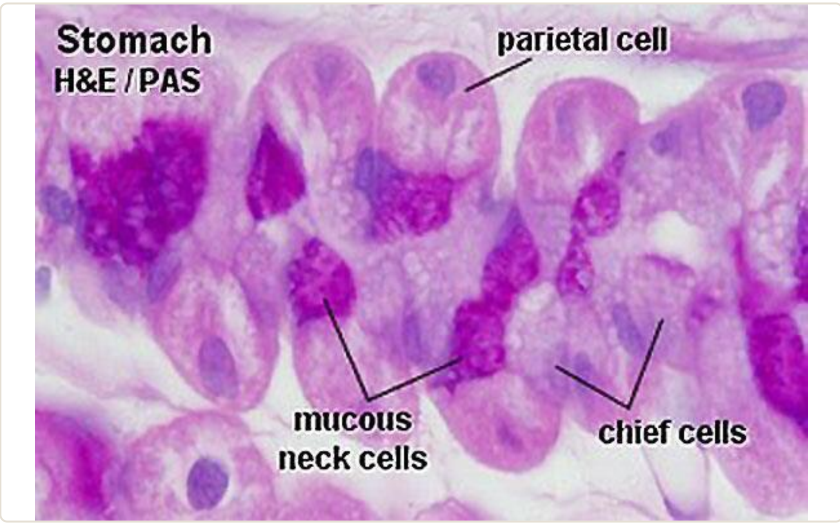
CELL	LOCATION	STRUCTURE	FUNCTION
Stem cell	Neck/isthmus	Low columnar, oval nuclei near cell base, high mitotic rate (turnover 4–7 days)	Regenerates gastric epithelium; proliferates when damaged (e.g. ulcer healing)
Surface mucous cell	Surface + pit lining	Tall columnar, basal nuclei, clear cytoplasm with mucin vacuoles	Secretes **neutral** mucus — protects surface from acid
Mucous neck cell	Neck, singly or in clusters between parietal cells	Irregular shape, basal nucleus, apical secretory granules	Secretes **less alkaline** mucus than surface cells
Parietal (oxyntic) cell	Neck (many), base (few)	Large, pale, round-to-pyramidal, 1–2 central nuclei, acidophilic cytoplasm, packed mitochondria	**HCl production** + intrinsic factor (glycoprotein needed for vitamin B12 absorption)
Chief (zymogenic) cell	Predominant at base	Smaller than parietal, basophilic, abundant RER/ribosomes, pepsinogen granules	Secretes pepsinogen → pepsin (protein digestion) and gastric lipase
Enteroendocrine (APUD) cell	Base	Small, round, clear cytoplasm, membrane-bound neurosecretory granules	Secretes gastrin, endorphins, serotonin



Surface mucous cells lining the pit — tall columnar, clear apical mucin.



Mucous neck cells and parietal cells — irregular basophilic neck cells beside large pale parietal cells.



Parietal cells, mucous neck cells, and chief cells together — the classic "name all three" exam field.

Enteroendocrine cell beside chief cells at the gland base.

LOOK-ALIKE PAIR — PARIETAL VS CHIEF CELL

	PARIETAL CELL	CHIEF CELL
Colour on H&E	**Pink** (acidophilic) — packed mitochondria	**Blue/purple** (basophilic) — packed RER
Size/shape	Large, round-to-pyramidal	Smaller
Position	Mostly neck, few at base	Predominant at base
Product	HCl, intrinsic factor	Pepsinogen, gastric lipase

4 · Pyloric Region — the completing detail

- Deep pyloric pits, short pyloric glands — mucus-secreting cells dominate, chief cells are scarce.
- **G cells** release gastrin; **D cells** secrete somatostatin.
- The muscularis' middle circular layer thickens here to form the **pyloric sphincter**.
- Serosa here is thin, covered by mesothelium.

PEARLS — WHAT MUST SURVIVE IF EVERYTHING ELSE IS FORGOTTEN

- Fundus and body are **histologically identical** — don't try to separate them on a slide.
- Pit:gland ratio is the fast regional discriminator: cardia (short:short), fundus/body (short:long), pylorus (long:short).
- **Parietal cell = pink, HCl + intrinsic factor. Chief cell = blue/purple, pepsinogen + lipase.**
- Stem cells sit in the neck/isthmus and drive re-epithelialisation after ulceration.
- Enteroendocrine cells secrete gastrin, endorphins, and serotonin from the gland base.

Transfer-task MCQs (unseen field, not slide recall)

1. A biopsy shows shallow gastric pits, long straight glands, and numerous large pink cells with central nuclei and dense mitochondria. Which region, and which cell dominates the description?

A) Cardia — mucous cells B) Fundus/body — parietal cells C) Pylorus — G cells D) Cardia — chief cells

2. A field shows very deep, elongated gastric pits with short coiled glands, mucus-dominant, near the duodenal junction. What region is this?

A) Fundus B) Body C) Cardia D) Pylorus

3. A patient develops pernicious anemia due to loss of intrinsic factor. Which cell type has been destroyed, and what else does it normally secrete?

A) Chief cell — pepsinogen B) Parietal cell — HCl C) Enteroendocrine cell — gastrin D) Mucous neck cell — alkaline mucus

4. A basophilic cell at the base of a gastric gland is packed with rough endoplasmic reticulum and membrane-limited zymogen granules. What does it secrete?

A) HCl and intrinsic factor B) Neutral surface mucus C) Pepsinogen and gastric lipase D) Gastrin

5. After a mucosal injury, which cell population proliferates to regenerate the gastric epithelium, and where is it located?

A) Chief cells, gland base B) Stem cells, neck/isthmus C) Parietal cells, neck D) Enteroendocrine cells, base

ANSWERS

1. B — shallow pits with long glands and abundant pink parietal cells is fundus/body.
2. D — deep pits with short coiled glands near the duodenum is the pyloric region.
3. B — parietal cells make both HCl and intrinsic factor; their loss causes B12 malabsorption and pernicious anemia.
4. C — basophilic, RER-rich cells with zymogen granules at the gland base are chief cells, secreting pepsinogen and gastric lipase.
5. B — stem cells in the neck/isthmus have a high mitotic rate and regenerate the epithelium after injury.

WHAT WAS LEFT OUT OF THIS SUMMARY

The detailed ultrastructural description of surface mucous cell microvilli and gap junctions was trimmed to the single line needed for the structure-function table — borderline cut, add back if Dr Hero stressed the junctional detail verbally. The rugae (mucosal folding when the stomach is empty) were mentioned only in passing rather than given their own section, since they're a gross/functional point rather than a histology-identification one.

Day 1 | Day 3 | Day 7 | Day 21 | Pre-exam